Exact Braid Algebra and Diamond-Norm Bounds for a Phenomenological Noisy Fibonacci-Anyon Qubit

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Abstract

We give a finite-dimensional, reproducible model of three Fibonacci anyons constrained to total charge τ. The braid operators are represented in the two-dimensional fusion-path basis using the standard F- and R-symbols. An explicit polynomial certificate proves the Yang–Baxter relation once two elementary scalar identities are supplied. We then define a physically motivated *effective* quantum-noise model incorporating fusion-basis phase flips and flagged erasure from the logical space, and prove an *exact* single-gate completely bounded trace-norm formula $\|\mathcal{E}_{q,\ell,U} - \mathcal{V}_U\|_\diamond = 2[\ell + (1-\ell)q]$. More generally, an exact closed form is given for arbitrary stochastic logical Pauli errors with flagged leakage. A telescoping theorem supplies circuit-level upper bounds under compatible channel composition. The accompanying Lean sources formalize the fusion rules, polynomial braid certificate, golden-ratio F identity, and abstract norm-composition bounds; the concrete complex-phase specialization and the superoperator diamond-norm instance remain proof obligations. No kernel-verification claim is made without a successful Lean build.

1 Scope, notation, and physical assumptions

Fibonacci anyons form a unitary braided fusion category with simple charges $\{\mathbf{1}, \tau\}$ and fusion rule

$$\mathbf{1} \otimes x = x, \qquad \tau \otimes \tau = \mathbf{1} \oplus \tau. \tag{1}$$

The resulting fusion-path spaces grow with Fibonacci numbers, a standard setting for topological quantum computation [1, 2, 3]. The construction below concerns a *single encoded qubit*, not a microscopic anyon Hamiltonian, a complete modular tensor category in Lean, or a demonstration of hardware-level fault tolerance.

We work with linear maps between matrix algebras and take the unhalved diamond norm $\|T\|_\diamond = \|T\|_{\mathrm{cb},1}$. Thus the operationally normalized channel error is $\varepsilon(T,S) = \frac{1}{2}\|T - S\|_\diamond$.

2 Fusion paths and the recoupling matrix

For three τ charges constrained to total τ, the two computational basis states correspond to intermediate charges $\mathbf{1}$ and τ. Put

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad u = \varphi^{-1} = \frac{\sqrt{5}-1}{2}, \qquad v = \varphi^{-1/2} = \sqrt{u}, \tag{2}$$

and define

$$F = \begin{pmatrix} u & v \\ v & -u \end{pmatrix}. \tag{3}$$

The identity $u^2 + u = 1$ and the choice $v^2 = u$ imply $F^2 = I_2$ and $F = F^\dagger$. In particular, F is unitary. The labels of the two basis vectors have precise topological meaning as fusion channels; the matrix entries are not arbitrary free qubit parameters.

Let $c_n(\mathbf{1}), c_n(\tau)$ denote the number of paths obtained by adjoining n copies of τ to the vacuum. With $(c_0(\mathbf{1}), c_0(\tau)) = (1, 0)$, fusion gives

$$\begin{pmatrix} c_{n+1}(\mathbf{1}) \\ c_{n+1}(\tau) \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} c_n(\mathbf{1}) \\ c_n(\tau) \end{pmatrix}. \tag{4}$$

Consequently $c_3(\tau) = 2$ and $c_n(\tau)$ obeys the Fibonacci recurrence.

3 Braiding and an exact polynomial certificate

We adopt the chiral convention

$$a = R^{\tau\tau}_{\mathbf{1}} = e^{-4\pi i/5}, \qquad b = R^{\tau\tau}_{\tau} = e^{3\pi i/5}, \qquad R = \mathrm{diag}(a, b). \tag{5}$$

The opposite chirality complex-conjugates both phases. The elementary braid generators are

$$B_1 = R, \qquad B_2 = FRF^{-1} = FRF. \tag{6}$$

As products of unitary matrices, both are unitary.

Theorem 3.1 (Three-strand braid relation)**.** *The matrices in* (6) *obey* $B_1B_2B_1 = B_2B_1B_2$.

Proof. For symbolic parameters u, v, a, b, set $H = u^2 + v^2 - 1$ and $Q = u^2(a - b)^2 + ab$. Direct expansion of $D = R(FRF)R - (FRF)R(FRF)$ yields the explicit certificate

$$D = \begin{pmatrix} (a-b)v^2Q - a(a^2u^2 + b^2v^2)H & -uv(a-b)Q - 2abuv(a-b)H \\ -uv(a-b)Q - 2abuv(a-b)H & -(a-b)v^2Q - b(a^2v^2 + b^2u^2)H \end{pmatrix}. \tag{7}$$

We already have $H = 0$. Put $z = b/a = e^{-3\pi i/5}$. Because $2\cos(3\pi/5) = -u$, one has $z^2 + uz + 1 = 0$. From $u^2 + u = 1$ it follows that $u^3 + 2u^2 = 1$; hence

$$u^2(1 - z)^2 + z = u^2(z^2 - 2z + 1) + z = (1 - u^2(u + 2))z = 0.$$

Multiplication by a^2 gives $Q = u^2(a - b)^2 + ab = 0$. Substituting $H = Q = 0$ in (7) proves $D = 0$. □

Remark 3.2. The symbolic identity (7) is checked exactly by SymPy and encoded as a conditional theorem in `Fibonacci/BraidAlgebra.lean`. The physical-phase lemma $Q = 0$, while proved above on paper, has *not* yet been discharged in the supplied Lean source. The analytic claim and Lean-kernel certification therefore have different verification statuses.

4 Effective dephasing and charge-leakage model

Let $\mathcal{H}_C = \mathbb{C}^2$, $\mathcal{H}_O = \mathbb{C}^2 \oplus \mathbb{C}|L\rangle$, and $J : \mathcal{H}_C \hookrightarrow \mathcal{H}_O$ be the canonical embedding. The ideal braid channel for a unitary $U \in \{B_1, B_2\}$ (or an arbitrary braid word) is

$$\mathcal{V}_U(\rho) = JU\rho U^\dagger J^\dagger. \tag{8}$$

We define two error parameters: $q \in [0, 1]$ is a stochastic logical fusion-basis Z-flip after the braid, and $\ell \in [0, 1]$ is an orthogonally flagged erasure/leakage event. With $Z = \mathrm{diag}(1, -1)$, define four 3×2 Kraus operators

$$K_0 = \sqrt{(1-\ell)(1-q)}\, JU, \qquad K_1 = \sqrt{(1-\ell)q}\, JZU, \tag{9}$$

$$K_2 = \sqrt{\ell}\, |L\rangle\langle 0|, \qquad K_3 = \sqrt{\ell}\, |L\rangle\langle 1|. \tag{10}$$

The noisy channel is $\mathcal{E}_{q,\ell,U}(\rho) = \sum_{j=0}^{3} K_j \rho K_j^\dagger$. This is completely positive, and

$$\sum_{j=0}^{3} K_j^\dagger K_j = ((1-\ell)(1-q) + (1-\ell)q + \ell) I_2 = I_2,$$

so it is trace-preserving. The orthogonal flag is a mathematical model for detected leakage from the computational fusion sector. This is *phenomenological*: q and ℓ are not inferred from a microscopic anyon bath, diffusion dynamics, or a measured hardware error rate. In particular, logical Z dephasing is not claimed to be a generic local topologically protected perturbation.

5 An exact diamond-norm expression

Theorem 5.1 (Exact one-gate error for flagged leakage and dephasing)**.** *For any unitary U on $\mathcal{H}_C$ and $q, \ell \in [0,1]$, the channels above satisfy*

$$\boxed{\|\mathcal{E}_{q,\ell,U} - \mathcal{V}_U\|_\diamond = 2\big[\ell + (1-\ell)q\big].} \tag{11}$$

Equivalently, the half-diamond error is $\varepsilon = \ell + (1-\ell)q$.

Proof. Write $t = \ell + (1-\ell)q$ and let $\mathcal{V}_{ZU}(\rho) = JZU\rho U^\dagger Z J^\dagger$, $\mathcal{L}(\rho) = \mathrm{Tr}(\rho)|L\rangle\langle L|$. Then

$$\mathcal{E}_{q,\ell,U} - \mathcal{V}_U = -t\mathcal{V}_U + (1-\ell)q\,\mathcal{V}_{ZU} + \ell\mathcal{L}. \tag{12}$$

Each of $\mathcal{V}_U, \mathcal{V}_{ZU}, \mathcal{L}$ is a quantum channel and therefore has diamond norm exactly one. The triangle inequality gives $\|\mathcal{E} - \mathcal{V}_U\|_\diamond \le t + (1-\ell)q + \ell = 2t$.

For a matching lower bound choose the legitimate density matrix $\rho_* = U^\dagger|+\rangle\langle+|U$, where $|\pm\rangle = (|0\rangle \pm |1\rangle)/\sqrt{2}$. The ideal output is $|+\rangle\langle+|$, while the noisy output is $(1-\ell)(1-q)|+\rangle\langle+| + (1-\ell)q|-\rangle\langle-| + \ell|L\rangle\langle L|$. In the orthonormal basis $\{|+\rangle, |-\rangle, |L\rangle\}$, their difference is diagonal with entries $-t, (1-\ell)q, \ell$. Its trace norm is $t + (1-\ell)q + \ell = 2t$. Since the diamond norm dominates the induced trace norm on any particular density matrix, it is at least $2t$, completing the proof. □

Remark 5.2. This is an exact expression for a channel $M_2(\mathbb{C}) \to M_3(\mathbb{C})$. It should *not* be substituted uncritically into a repeated-gate circuit on $M_2(\mathbb{C})$: after leakage, the output space has changed. A valid circuit composition must include a compatible propagation or recovery rule for the flag.

6 Extension: arbitrary stochastic Pauli noise

The exact formula is not restricted to dephasing. Let $P_0 = I$, $P_1 = X$, $P_2 = Y$, $P_3 = Z$, and $p_\alpha \ge 0$ with $\sum_{\alpha=0}^{3} p_\alpha = 1$. Define a channel $\mathcal{E}_{\boldsymbol{p},\ell,U} : M_2(\mathbb{C}) \to M_3(\mathbb{C})$ by Kraus operators

$$K_\alpha = \sqrt{(1-\ell)p_\alpha}\, JP_\alpha U \quad (0 \le \alpha \le 3), \qquad K_4 = \sqrt{\ell}\,|L\rangle\langle 0|, \quad K_5 = \sqrt{\ell}\,|L\rangle\langle 1|.$$

Theorem 6.1 (Exact Pauli-plus-erasure diamond distance)**.** *For any unitary U, any probability vector $\boldsymbol{p}$, and $0 \le \ell \le 1$,*

$$\boxed{\|\mathcal{E}_{\boldsymbol{p},\ell,U} - \mathcal{V}_U\|_\diamond = 2\big[1 - (1-\ell)p_0\big].} \tag{13}$$

Proof. The ideal Pauli branch occurs with probability $(1-\ell)p_0$; all other branches carry nonnegative weights summing to $t = 1 - (1-\ell)p_0$. Subtracting $\mathcal{V}_U$ leaves the ideal channel with coefficient $-t$, plus CPTP Pauli branches whose weights sum to $(1-\ell)(1-p_0)$ and an erasure channel of weight ℓ. Triangle inequality bounds the diamond norm by $2t$.

For a matching lower bound, take an auxiliary qubit and the normalized Bell state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$, with input $(U^\dagger \otimes I)|\Phi^+\rangle$. The four states $(JP_\alpha \otimes I)|\Phi^+\rangle$ are orthonormal; any erasure output lies in the orthogonal flagged sector. Therefore the difference between output density operators has negative eigenvalue $-t$ on the ideal Bell state, positive eigenvalues $(1-\ell)p_\alpha$ for $\alpha \ne 0$, and positive flagged eigenvalues summing to ℓ. Its trace norm is exactly $2t$. □

Remark 6.2. Equation (11) is the special case $p_0 = 1 - q$, $p_3 = q$, $p_1 = p_2 = 0$. Unlike that special case, the general Pauli lower bound uses a reference qubit. The result is a direct analytic proof, not a separately solved numerical SDP and not yet a kernel-formalized completely bounded norm theorem.

7 Compositional bounds for braid circuits

Theorem 7.1 (Telescoping diamond bound)**.** *Suppose $\mathcal{E}_j$ and $\mathcal{U}_j$ are CPTP maps with compatible input/output spaces for $j = 1, \ldots, n$. Then*

$$\|\mathcal{E}_n \circ \cdots \circ \mathcal{E}_1 - \mathcal{U}_n \circ \cdots \circ \mathcal{U}_1\|_\diamond \leq \sum_{j=1}^{n} \|\mathcal{E}_j - \mathcal{U}_j\|_\diamond. \tag{14}$$

Proof. Insert and subtract intermediate products so that each summand differs at only a single gate. The diamond norm obeys the triangle inequality and is submultiplicative under composition. Every CPTP map has diamond norm one. Bounding each summand thus gives (14). □

If compatible extended-space channels are built and each is separately proved to satisfy $\delta_j \leq 2t_j$, then the operationally normalized circuit error satisfies

$$\varepsilon_{\text{circuit}} \leq \min\{1, \sum_{j=1}^{n} t_j\}. \tag{15}$$

Any additional braid-compilation error relative to a target logical gate must be included by the triangle inequality. The results do not establish a topological threshold or a suppression law for q or ℓ as a function of quasiparticle separation or temperature.

8 Lean sources and scope of formal verification

The accompanying source files implement five proof layers:

1. `Fusion.lean`: fusion outcomes, path recurrence, and the two-dimensional three-anyon fusion basis.
2. `BraidAlgebra.lean`: explicit 2×2 matrix multiplication and a polynomial certificate for the braid relation under $H = Q = 0$.
3. `GoldenF.lean`: the golden-ratio F specialization and the physical exponential phases; *the exact physical-phase identity is an explicit theorem hypothesis.*
4. `QuantumNoise.lean`: explicit three-level Kraus operators for dephasing and flagged leakage; normalization of the scalar weights.
5. `DiamondComposition`, `NoiseCoefficients`, `PauliWeights`: normed-ring telescoping bounds and scalar Pauli-weight identities. A concrete mathematical instance identifying this norm with the completely bounded trace norm on superoperators is *not supplied.*

The exact quantum-channel theorems 5.1 and 6.1 are proved above by finite-dimensional functional analysis, but neither has yet been formalized in Lean. The accompanying symbolic and numerical tests are executable using the supplied Python script. The Mac audit script requires a working installation of the pinned Lean toolchain to determine whether the Lean files type-check; no completed Lean build or kernel-axiom audit is included with these materials. The audit prints both complete theorem statements and kernel axiom dependencies; a successful build alone is not treated as proof of the intended scientific claims.

9 Novelty and limitations

The Fibonacci braid representation and universality are known results [1, 2, 3]. Diamond-norm SDP methods are well-established [4, 5], and phase damping is a standard effective noise mechanism [6].

Experimental digital demonstrations of Fibonacci-anyonic braiding exist [7]. The single-gate formula (11) is an elementary but useful closed-form property of the explicit phenomenological flagged channel. No claim of world priority or new physical topological protection is made.

For complete machine-checked verification, further steps include: (i) kernel proof of the exact physical complex-phase identity and possibly relevant hexagon/pentagon equations; (ii) a concrete CPTP and Kraus-completeness proof for the matrix channel in Lean; (iii) a trace-norm / diamond-norm formalization yielding theorem 5.1 without abstract assumptions; and (iv) a fully specified composable leakage dynamics, with verified hardware calibration if physical predictive claims are intended.

A Reproducibility checklist

From the extracted project root with a network-enabled Lean 4.19.0 installation and Mathlib, run

```
python3 scripts/check_numeric.py
zsh scripts/run_lean_audit_mac.zsh
```

Inspect `results/numerical_report.json` and `results/lean_axioms.log`. The second output is generated *only if a real Lean run takes place.* Review `#check` statements in `Fibonacci/Audit.lean`; a theorem whose conclusion is merely `True` would not certify positivity or optimality. The supplied sources contain no `sorry` placeholders, but *the absence of such tokens does not certify uncompiled source.*

References

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